

DRM-ETP: A Dynamic Rate Matching-Based Entanglement Transport Protocol in Quantum Networks

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Abstract—The entanglement transport protocol with a connection-oriented mode ensures the reliable distribution of remote entanglement by reserving dedicated resources on the selected path for users in a quantum network. In most existing protocols, entanglement generation and resource allocation operate with the support of global network-synchronized time slot. However, such synchronization in a large-scale quantum network is challenging, and the idealized time slot model is not conducive to continuous and concurrent requests. Meanwhile, different link performance in memory capacity and entanglement generation rate brings out critical issues, such as long distribution delay and low resource utilization, which has not been adequately addressed by the existing protocols relying on a heuristic adoption of TCP-like transport modes. In light of these observations, we propose a dynamic *rate matching*-based entanglement transport protocol called DRM-ETP, which allocates different memory units on each link along an entanglement distribution path. Moreover, DRM-ETP incorporates periodic forward and backward interactions to implement fine-grained feedback and a dynamic memory allocation based on priority differentiation. These mechanisms mitigate congestion and unfairness arising from resource contention among burst requests on shared links. Extensive simulation results demonstrate that DRM-ETP significantly outperforms the existing protocols in terms of throughput and resource utilization, with less distribution delay and higher fidelity. Moreover, DRM-ETP exhibits rapid and fair convergence when handling burst requests. Our study opens up possibilities for deploying efficient entanglement transport in quantum networks, thereby holding the promise of enhanced compatibility and novel functionality.

Index Terms—Entanglement distribution, rate matching, entanglement transport protocol.

I. INTRODUCTION

RECENTLY, quantum networks, characterized by distinctive quantum properties, have gained prominence as a focal point of research, catalyzing advancements in the application of cutting-edge quantum information technology [1], [2], [3]. Due to the unique features of quantum systems, entanglement distribution between distant communication users is one of the essential building blocks of quantum networks, which allows quantum nodes to interconnect and interact with each other [4]. Leveraging short-haul entanglement shared by adjacent nodes to construct end-to-end (E2E) entanglement through swapping on a selected path becomes a prerequisite for unlocking the potential of quantum applications, such as quantum key distribution [5], [6] and quantum computing [7].

To enhance the flexibility and autonomy of constructing quantum networks, researchers propose a multi-layer protocol stack for quantum networks [8], [9], [10]. This includes the physical layer [11], link layer [12], network layer [13], and transport layer [14], application layer, progressing from the bottom to the top. Among them, efficient and controllable long-distance entanglement transport stands out as a key driver for enhancing distribution performance, serving as the primary design objective at the transport layer [10]. Analogous to the virtual circuit mechanism in classical networks, an entanglement transport protocol adopts a connection-oriented approach to enhance reliability. This implies the allocation of dedicated entanglements or memory units along the reserved path for a connection [15]. To date, some studies [16], [17], [18] concentrate on connection-oriented transport modes, yet these designs remain imperfect and have not fully accounted for some quantum properties (e.g., neglect of resource competition and quantum decoherence). Given the uniqueness of the entanglement transport mechanism and enabling technology, new challenges, surface in the design of entanglement transport protocol, distinguishing it significantly from classical techniques.

The first challenge is the short timeliness of entanglement. Inherent quantum decoherence characteristic [19] imposes

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constraints on the survival time of entangled pairs for processing. However, the existing solutions [20], [21], [22] resort to a global network-synchronized time slot to achieve the generation and allocation of entanglement resources rather than distributed local decision. Especially in the future large-scale quantum networks, the synchronization overhead and decoherence side effects caused by non-negligible global scheduling are important issues that the current protocol has not fully resolved. Furthermore, these schemes for implementing batch processing requests based on synchronous mode are often not suitable for serving some applications with high real-time performance requirements, which may reduce the flexibility of network services. Therefore, a transport protocol with lower time cost and better real-time performance should support an asynchronous and continuous mode to reduce distribution delay.

Another challenge involves non-optimal distribution efficiency, primarily stemming from different link performance. In quantum networks, a high-quality E2E entanglement connection is initially established. Subsequent data transport relies on teleportation [23], [24], as opposed to the direct hop-by-hop data transport in classical networks. The entanglement transport protocol requires the coordinated cooperation of all intermediate nodes on the selected path to achieve efficient entanglement distribution. Specifically, all intermediate nodes are intended to swap entangled pairs shared between adjacent nodes in parallel. To distribute E2E entanglement with less delay, it is essential to maintain a consistent entanglement generation rate on each link along the entanglement distribution path. In this manner, each entangled pair can be swapped immediately upon generation, without waiting time and resource wastage. However, the existence of different link performance, which is defined as links on the selected path differ in entanglement generation rate and memory capacity, complicates the proprietary resource reservation. On quantum links with different performance along a path, the implementation of optimal resource allocation schemes based on *rate matching* significantly enhances transport efficiency.

To tackle the aforementioned challenges, firstly, we adopt the connection-oriented entanglement transport protocol generally designed by reserving dedicated memory units on each link along a path. Besides, protocol should runs in an asynchronous mode with abandoning the time slot setting of full-network synchronization. Secondly, to enhance transport performance further, we introduce a novel mechanism called *rate matching*, which allocates different memory units on each link, taking different link performance into account. Based on these mechanisms, we develop a connection-oriented continuous entanglement transport protocol. Specifically, our protocol gathers path status information through forward propagation and allocates memory units through backward propagation in a periodic manner. Then, dynamic memory allocation based on an explicit average threshold is incorporated to handle resource contention resulting from burst requests. The primary contributions of this paper are summarized as follows:

- We design a novel *rate matching* mechanism in connection-oriented and continuous processing transport mode. *Rate matching* addresses the critical challenges

related to distribution delay and different link performance by reserving different memory units on each link along the designated path.

- We propose a dynamic *rate matching*-based entanglement transport protocol (DRM-ETP) in quantum networks. DRM-ETP facilitates fine-grained path state updates through periodic forward and backward interactions, employing a dynamic memory allocation to mitigate congestion and ensure fairness in handling resource contention from burst requests on shared links.
- We conduct extensive simulations on selected paths with varying lengths and random network topology. DRM-ETP demonstrates superior performance compared to the existing transport protocols, exhibiting notable advantages in distribution delay, resource utilization, and throughput with accelerated convergence.

The rest of this paper is organized as follows. We list and discuss the related work in Section II. We introduce the system model and our motivation in Section III. Section IV presents the protocol design and analysis in detail. Then, in Section V, we perform extensive simulations to demonstrate the effectiveness of our protocol compared with the existing protocols. Finally, we conclude the work in Section VI.

II. RELATED WORK

Up to now, many studies attempt to improve the performance of entanglement distribution from the perspective of protocol stack for quantum networks. Similarly to classical network, multi-layer protocol stack [9], [10], [25] of quantum networks is proposed. Specifically, it has a physical layer [26], [27] to generate entanglements and deliver qubits by using advanced devices and technology. The function of the link layer [12] is to robustly generate short-distance quantum entanglement between two adjacent nodes. Ulteriorly, based on the abstraction of underlying technologies, many studies are conducted at higher layers.

Above the link layer, there is a network layer focusing on routing [28] and resource allocation to distribute entangled pairs for quantum nodes that are far away from each other. Van Meter et al. [28] studied applying the Dijkstra algorithm by defining a new link cost to find a single path for each source-destination (SD) pair in quantum networks. Pant et al. [21] designed a greedy algorithm in grid networks, which uses a group of paths with the fewest hops to establish entangled pairs, based on the information about all the successfully generated entangled pairs. Shi and Qian [20] proposed the extended Dijkstra's algorithm to find the most reliable paths by considering the success probability of generating entangled pairs for one of the SD pairs and trying to establish as many entangled pairs as possible along these paths. Li et al. [22] proposed three algorithms (PS/PF/PU) for the scheduling of capacity allocation on the edges of the quantum network for multiple requests based on successfully generated entangled resources. The network layer protocols currently focus more on connectivity and optimizing resource allocation for multiple requests and pay little attention to improving transport performance on selected paths.

Reliable entanglement distribution is currently mostly based on connection-oriented transport features. Typically, Li et al. [16] proposed a pioneering connection-oriented protocol to provide reliable and stable entanglement distribution according to the requirement of quantum applications. Kozłowski et al. [13] designed a quantum data plane network protocol for creating long-distance E2E entangled pairs. This protocol is efficient by utilizing virtual circuits, building upon a robust link-layer protocol. These protocols provide a basic version for studying connection-oriented transport protocols, but common issues such as conflicts and congestion during transport have not been thoroughly resolved, and transport efficiency also needs to be improved.

In recent research on entanglement transport, Zhao et al. [14] proposed Tele-DTP, a pioneering distributed entanglement transport protocol using teleportation. Tele-DTP aims to establish a connection by announcing the same number of memory units on all links along a reserved path and using TCP-like control methods to solve congestion. However, Tele-DTP operates in a synchronous, time-slotted fashion and cannot guarantee that the allocated memory units will be fully filled with entanglement due to the probability of generation within a time slot. Furthermore, in Tele-DTP, the actual entanglement generation rate of each link is inconsistent due to different link performance, leading to a mismatch phenomenon of generation rates along the selected path. For instance, high-rate links provide redundant resources, while low-rate links are in short supply. Consequently, more resources are consumed without effectively distributing additional available E2E entangled pairs. Therefore, announcing the same memory units along the reserved path is not applicable in quantum networks with different generation rates and memory capacities. There exists the potential for further enhancing the transport performance of Tele-DTP.

In summary, most of the related works run with the support of the global network-synchronized time slot, which is challenging to deploy practically and incompatible with quantum properties, especially in a large-scale network. In addition, the existing transport protocols are inspired by the classical TCP protocol. Their performance and efficiency cannot be reliably guaranteed due to a lack of consideration for different link performance. To solve these problems, referring to the core function of the transport layer in a typical quantum network protocol stack, a transport protocol is expected to precisely coordinate the resource allocation of each link on the selected path, solve the inefficiency problem caused by different link performance. In addition, the entanglement transport protocol should operate in continuous and asynchronous quantum networks and abandon the synchronous time slot assumption to enhance compatibility and usability.

III. SYSTEM MODEL AND MOTIVATION

A. Parallel Entanglement Swapping

Distributing entanglements among remote quantum nodes is one of the essential building blocks for achieving advanced quantum applications. However, it is difficult to enable two

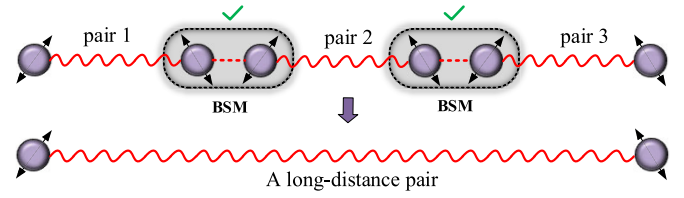


Fig. 1. Extend three short pairs to a long pair by parallel entanglement swapping.

remote quantum nodes to share entangled pairs directly due to the inherent photon loss and noise of quantum channels. Entanglement swapping [29] is an essential operation used to effectively extend the distance of entanglement. To diminish the distribution delay, similar to most studies [13], [20], [22], the selected path can adopt a parallel manner in which all intermediate nodes perform Bell-state measurement (BSM) [30] operations to swap entanglement simultaneously. The end user establishes entanglement with another user based on the convergent measurement results. As shown in Fig. 1, three entangled pairs shared by three pairs of adjacent nodes are extended to be long-distance entanglement through two successful BSM operations. Unfortunately, due to the probabilistic nature of quantum measurement [31], the influence of decoherence on entangled bits [32], and the limitations of the development of physical devices [33], entanglement swapping always shows a probabilistic success. Therefore, the probability of E2E entanglement distribution reduces with the more BSMs, i.e., more repeaters.

B. Transport Mode and Burst Requests

Transport modes commonly used in classical networks, such as connection-oriented and connectionless, can serve as references in quantum protocol design. A connectionless protocol [34], [35] swaps entanglement hop-by-hop, and its entanglement transport does not necessitate pre-established resource reservation. Consequently, a connectionless protocol is easily deployed but lacks performance assurance, especially in large-scale networks. In contrast, the so-called connection [16] refers to a dedicated E2E entanglement distribution process established between requests. A connection requires a reliable path, which is usually determined by a routing algorithm, and reserves dedicated network resources on this path. In this paper, it is assumed that a request initiates an entanglement distribution connection. During the connection period, entangled pairs are continuously attempted to be generated, and each repeater performs entanglement swapping to extend the entanglement distance. When the request is satisfied (e.g., enough entangled pairs have been distributed), the source node initiates connection revocation to release the reserved network resources on each node on the path. In our design, due to the continuous generation and consumption of entanglement, a connection needs to occupy dedicated memory units on each node of the path to exclusively utilize entanglement resources. Specifically, the entangled pairs generated in the memory units reserved for a certain connection are considered as dedicated resources belonging to that connection. Other

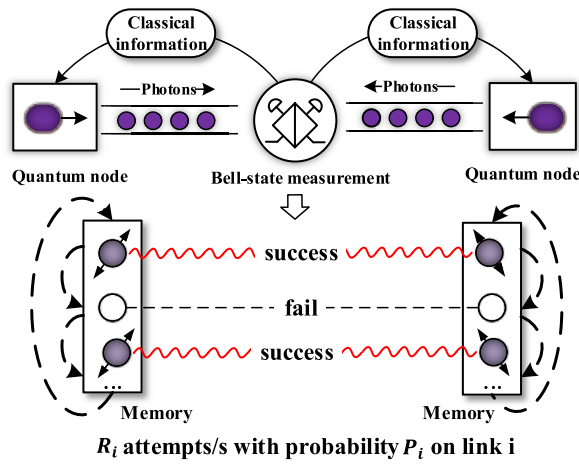


Fig. 2. Continuous entanglement generation process.

connections, although they may also pass through the same node, have no right to use them.

Since the quantum network operates in an asynchronous and continuous mode, new requests may occur at any time, and we define this characteristic as “burst”. In other words, a “burst request” is the same as initiating a new connection from new users. For better deployment compatibility, our protocol provides real-time performance guarantees for new burst requests instead of waiting to collect a batch of requests and then process them uniformly.

C. Entanglement Generation Model

From the perspective of E2E entanglement transmission, it is essential to abstract and quantify the entanglement generation process without focusing on the specific details of the transmission medium or channel type. As illustrated in Fig. 2, similar to many existing studies [22], [36], we assume that quantum link i has a maximum and stable generation attempt rate, denoted as R_i . We also abstract and quantify adverse effects, such as inaccurate measurement and channel noise, as the successful generation probability P_i . Specifically, if two adjacent nodes on a link each have C_i quantum memory units, then each memory unit will sequentially perform R_i/C_i entanglement generation attempts per unit time. Upon a successful attempt, the entanglement is retained in the corresponding memory for subsequent use; conversely, if the attempt fails, memory units will wait for the next generation attempt.

Furthermore, since our protocol operates in a continuous processing network without time slot assumptions, entangled pairs are continuously generated and consumed on each link. Successfully generated entangled pairs are stored in the corresponding memory units for subsequent operations. Therefore, if a request connection occupies more memory units at the end nodes of one link, then a better entanglement generation performance will be allocated correspondingly.

D. Different Link Performance

The different link performance in quantum networks mainly comes from two aspects. 1) Heterogeneous network structure.

When constructing a quantum network, we can use various communication media (e.g., optical fibers or free space) to construct quantum links to achieve entanglement distribution, such as a ground-based and satellite-based hybrid quantum network [3]. This heterogeneity may lead to significant differences in link performance, which can be analogized to the difference in communication performance between wireless and optical fibers in classical networks. 2) Performance differences in entanglement generation and storage. The entanglement generation process entails the entanglement preparation source preparing entangled particles. These particles are then distributed to the end nodes of the link through a quantum channel and stored in quantum memory for subsequent use [37]. High-performance entanglement generation sources possess a higher entanglement preparation attempt rate. The success probability of distributing entanglement between nodes at both ends of a link declines exponentially with physical distance [38], [39]. Additionally, entangled pairs stored in high-capacity and more advanced quantum memories have a longer lifespan to resist decoherence. Hence, even in homogeneous networks, factors such as channel distance, loss and noise characteristics, and storage technology play a crucial role in determining link performance.

E. Motivation

Here we demonstrate the research motivation by combining the two challenges mentioned earlier.

1. Short timeliness of entanglement. Entangled pairs are fragile and highly susceptible to decoherence and inefficiency. In advanced studies, the effective storage time is currently approximately from nanoseconds to milliseconds [40], depending on the specific entanglement preparation techniques and storage medium, such as the DLCZ scheme [41], rare-earth-ion-doped crystals [42], quantum dots [43], and nitrogen-vacancy (NV) centers [44]. The short lifetime poses a challenge to the entanglement routing and transport scheme based on synchronous time slots. The currently widely used time synchronization protocols on the Internet, such as the network time protocol (NTP) and precision time protocol (PTP), lack sufficient time synchronization accuracy. For example, the accuracy on a wide area network is several tens of milliseconds. In the latest research, Bush et al. [45] proposed high-precision time synchronization mechanisms for quantum networks, but still cannot effectively solve the non-negligible time synchronization overhead. Furthermore, implementing batch processing requests based on synchronous mode is not suitable for serving some applications with high real-time performance requirements, which reduces the flexibility of network services. Therefore, a transport protocol with lower time cost and better real-time performance should support an asynchronous and continuous mode to reduce distribution delay.

2. Non-optimal distribution efficiency. On an entanglement distribution path, a crucial question is how many memory units should be allocated on each link. The existing transport protocol, Tele-DTP, provides a possible solution by occupying the same number of memory units. We illustrate a reserved path with three links (Alice-N1, N1-N2, N2-Bob)

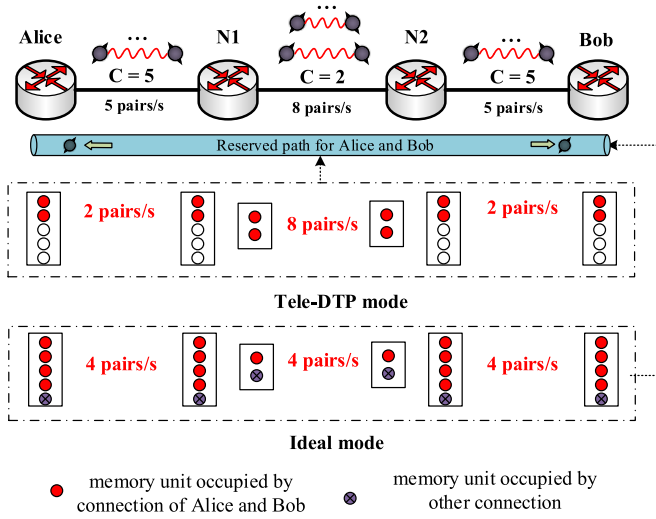


Fig. 3. Performance differences caused by two memory allocation modes. Tele-DTP always allocates the same memory units, while the ideal mode achieves *rate matching* by occupying different numbers of memory units.

between Alice and Bob by taking Fig. 3 as an example when considering the different link performance. The memory capacity and the entanglement generation rate of the links differ from each other. Tele-DTP declares the same 2 memory units on each link, resulting in mismatched entanglement generation rates (2 pairs/s, 8 pairs/s, and 2 pairs/s, respectively). This inconsistency leads to undesirable consequences: in the high-rate link N1-N2, entanglement supply exceeds demand, and most additional resources are wasted, while in the low-rate links Alice-N1 and N2-Bob, entanglement demand exceeds supply, hindering further performance improvement. The ideal allocation mode achieves *rate matching* (4 pairs/s) by reserving different amounts of memory units for each link on the selected path (4 units, 1 unit, and 4 units, respectively). Statistically, resource generation and swapping between each link can be performed synchronously. In this theoretically optimal situation, once entanglement is generated, it will be immediately swapped, resulting in minimal waiting and waste for entangled pairs. Moreover, this mode also leaves spare memory units on each link that can be utilized by other connections.

IV. DRM-ETP PROTOCOL DESIGN

A. Overview

Based on the previous analysis, an overarching purpose of this paper is to design a novel transport protocol satisfying the following two goals:

Asynchronous and continuous running process. The entanglement transport protocol should be adaptively deployed in an asynchronous and continuous quantum network. Resource generation and consumption, scheduling decisions, connection establishment, and revocation occur at any time without relying on time slot assumptions or a centralized controller. Additionally, the protocol must address memory contention among multiple burst requests and ensure fairness between new and old requests to prevent congestion.

TABLE I
NOTATION LIST

Parameters	Description
C_i	The memory capacity on adjacent nodes of link i ;
O_i^j	Calculated allocated capacity of one connection of link i in j -th cycle;
$\hat{O}_i^j$	Actual allocated capacity of one connection of link i in j -th cycle;
M_i^j	Remaining spare memory units of link i in j -th cycle;
R_i	The maximum entanglement generation attempt rate on the link i ;
P_i	Entanglement generation probability of link i ;
K_i	Entanglement generation rate on one unit of link i , $K_i = R_i * P_i / C_i$;
Q	Entanglement swapping probability of each repeater;
N_i	Number of requests (connections) of the shared link i ;

Reliable and efficient entanglement distribution. The classical TCP protocol regulates the sending window through feedback mechanisms, such as timeouts, without involving the processing mechanism of the underlying protocol. However, the probabilistic nature of quantum operations poses a challenge: the feedback mechanism cannot reliably distinguish whether the decline in transport performance is due to network congestion or failed quantum operations. A feasible approach is to allocate exclusive memory units to ensure ownership of entangled pairs in the corresponding memory. In addition, the protocol needs to reserve different memory units on links with different link performance to achieve *rate matching* and improve the entanglement distribution efficiency.

Therefore, in this paper, the primary objective of our proposed dynamic *rate matching*-based entanglement transport protocol (DRM-ETP) is to operate in a connection-oriented and asynchronous manner. During the implementation process, DRM-ETP should address the following key issues.

- **How to achieve rate matching on the selected path?** The entanglement transport protocol considers the maximum available entanglement generation rate on the selected path and derives the respective allocated memory capacity based on this consistent rate.
- **How to handle the contention with burst requests?** Asynchronous operations inevitably lead to dynamic resource contention on shared links caused by burst requests, as well as resource reallocation after the revocation of requests. The transport protocol is based on a fair strategy of average allocation and introduces dynamic temporary allocation of spare resources.
- **How to reach fast convergence?** When handling new connection establishments and old connection revocations, the entanglement transport protocol should consider fine-grained feedback sent from each repeater to achieve rapid performance convergence. In our protocol, we design an active periodic forward and backward information feedback mechanism originating from end nodes, rather than triggering feedback on repeaters for compatibility and easy deployment.

For convenience, the key symbols and their definitions used in the following sections are shown in TABLE I.

B. Detailed Solutions

1) **Rate Matching**: From the perspective of upper-layer transport control, we adopt the entanglement generation model as described in Sec. III-C. For one reserved path (n links) $\mathcal{P} = \{\langle R_1, C_1, P_1, O_1 \rangle, \langle R_2, C_2, P_2, O_2 \rangle, \dots, \langle R_n, C_n, P_n, O_n \rangle\}$, we analyze and demonstrate the necessity and optimality of *rate matching*. Statistically, each entangled pair is filled into each memory unit at an average interval of $\frac{1}{R_i P_i}$. The average fill interval of each memory unit is $\frac{C_i}{R_i P_i}$. In one unit time Δt , the average number of special entangled pairs for this connection on link i is:

$$\lambda_i = O_i \times \frac{1}{\frac{C_i}{R_i P_i}} = \frac{O_i}{C_i} R_i P_i. \quad (1)$$

The number of generated entanglement follows the Poisson distribution [46], [47] due to the following two aspects. 1. The entanglement generation process on quantum link is a typical binomial distribution, which satisfies the independent event assumption. 2. The quantum link has a relatively stable entanglement generation attempt rate, which satisfies the assumption of a fixed average rate. Then in Δt , the probability of generating k entangled pairs on link i is:

$$P_i(N(t) = k) = \frac{(\lambda_i)^k e^{-\lambda_i}}{k!}. \quad (2)$$

The probability of generating at least one entangled pair on link i is:

$$\hat{P}_i = 1 - P_i(N(t) = 0) = 1 - e^{-\lambda_i} = 1 - e^{-\frac{O_i}{C_i} R_i P_i}. \quad (3)$$

Since the generation of each link is independent of each other, the probability that all links on the path generate at least one entangled pair is:

$$\begin{aligned} \hat{P} &= \prod_{i=1}^n \hat{P}_i = \prod_{i=1}^n (1 - P_i(N(t) = 0)) \\ &= \prod_{i=1}^n (1 - e^{-\frac{O_i}{C_i} R_i P_i}). \end{aligned} \quad (4)$$

We calculate the $\hat{P}$ of the two different allocation modes in one unit time in Fig. 3 as follows:

$$\hat{P}_1 = (1 - e^{-2})(1 - e^{-8})(1 - e^{-2}) = 0.747, \quad (5)$$

$$\hat{P}_2 = (1 - e^{-4})(1 - e^{-4})(1 - e^{-4}) = 0.946, \quad (6)$$

where $\hat{P}_1$ is for Tele-DTP mode and $\hat{P}_2$ is for ideal mode. Intuitively, in ideal mode, the synchronization between links on the path is better, which is beneficial to decrease the storage time after the entanglement generation, that is, to weaken the influence of decoherence and reduce the distribution delay.

To achieve *rate matching*, a consensus must be reached on all links in the selected path to determine the maximum entanglement generation rate assigned to a single request. Subsequently, this agreed-upon rate is translated into the occupied capacity of memory units on each link (i.e., between two adjacent quantum nodes). The entangled pairs successfully generated and stored in the exclusively allocated memory units can then be utilized for subsequent entanglement swapping for this connection. It's crucial to note that this matched rate is not

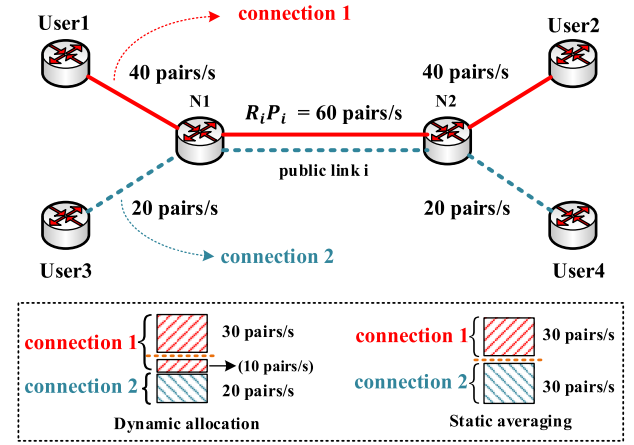


Fig. 4. Two allocation results on the shared link i lead to different distribution performance.

constant due to the dynamic nature of connections influenced by burst requests.

In addressing resource contention when burst requests access or exit shared links, two possible scenarios are identified: 1) Existing requests occupy partial or total memory units of the shared link. The allocation of memory units can be adjusted to accommodate new requests. 2) Existing requests are revoked, resulting in released memory units. The remaining requests should be reasonably and fairly reoccupied. Dealing with multi-request memory contention primarily involves allocating and adjusting memory units fairly and quickly when encountering burst requests. Specifically, we next design average threshold and dynamic memory allocation mechanisms to address these issues.

2) **Dynamic Allocation Based on Average Threshold**: When multiple requests N_i share a link i , a straightforward allocation method involves dividing the memory units C_i equally, denoted as $\frac{C_i}{N_i}$, which also is equivalent to the average rate $\frac{R_i P_i}{N_i}$. Rounding up is commonly applied to ensure integer values. While this method achieves explicit fairness in an idealized scenario, achieving *rate matching* on the path requires considering not only local contention but also congestion on other links, making the actual allocation results on a shared link more complex.

To overcome the performance limitations imposed by a static averaging threshold, we design the dynamic allocation mechanism which serves two purposes. First, connections can utilize free memory units above the threshold with lower priority, even if those units are intended for other connections. Second, some connections with higher priority can reclaim memory units below the threshold that are temporarily occupied by lower-priority connections. We use Fig. 4 as an illustrative example, where the average threshold of the shared link N1-N2 is 30 pairs/s. Strictly adhering to the averaging threshold would result in actual transport rates on the two connections being 30 pairs/s and 20 pairs/s, if swapping is always successful. However, since connection 2 cannot further increase the generation rate on the other two links, connection 1 can dynamically occupy redundant resources (10 pairs/s) on the shared link with higher priority. Consequently, the

actual transport rates on the two connections can increase to 40 pairs/s and 20 pairs/s. Dynamically allocating these redundant resources can bring more significant benefits when dealing with more connections.

C. Protocol Design

DRM-ETP is implemented on the determined path with n links by adopting a distributed processing mechanism for better scalability. To achieve the above functions and ensure the robustness of the protocol, DRM-ETP includes the following three processes: the *Forward Seeking Process*, the *Backward Allocation Process*, and the *Connection Revocation Process*. In addition, the first two processes are executed periodically to discover whether resource allocation needs to be readjusted on the path. We define the time interval between two adjacent forward search processes as one cycle. The length of this cycle is determined by the network status. For example, when the number and location of requests are relatively stable, the cycle can be lengthened. When there are more burst requests, the cycle can be shortened to adapt to the rapidly changing network status. After the request is satisfied, the *Connection Revocation Process* is executed to release the occupied memory units. In general, the explicit processing content of the $(j+1)$ -th cycle is derived based on the processing result of the j -th cycle. Then, the execution contents of the three processes are as follows:

1) *Forward Seeking Process*: In the $(j+1)$ -th forward cycle, source node initiates a connection and sends the seeking signal to destination hop by hop along the selected path. Each node along the path attaches the current data tuple of the forward link i to the classical packet which contains the following information:

$$[W_i^{j+1} = \max\{\text{threshold}_i, \hat{O}_i^j + M_i^{j+1}\}, K_i], \quad (7)$$

where $\text{threshold}_i = \frac{C_i}{N_i}$ means total memory capacity divided by the number of connections passing through this link, i.e., the average threshold used as the baseline. $\hat{O}_i^j$ represents the current number of memory units already occupied by this connection in the j -th cycle, and M_i^{j+1} denotes the number of spare memory units not yet occupied by any connection in this $(j+1)$ -th cycle. The current actual available capacity is the sum of the occupied capacity of the previous cycle and the current spare memory. Taking the maximum value reflects the dynamic allocation mechanism based on threshold_i . K_i represents the generation rate of one memory unit, which is used to calculate the final unified entanglement generation rate on destination.

$$K_i = R_i * P_i / C_i. \quad (8)$$

Furthermore, destination gathers this information from each link i and obtains their available rate as follows:

$$S_i^{j+1} = W_i^{j+1} * K_i. \quad (9)$$

By calculating the minimum value for satisfiability of all links, the final maximum matching rate is calculated as $S_{max}^{j+1} = \min\{S_1^{j+1}, S_2^{j+1}, \dots, S_n^{j+1}\}$.

Algorithm 1 Memory Management Algorithm

Input: Actual allocated capacity $\mathcal{O}^j = \{\hat{O}_1^j, \hat{O}_2^j, \dots, \hat{O}_n^j\}$ in the j -th cycle;

Output: Actual allocated capacity $\mathcal{O}^{j+1} = \{\hat{O}_1^{j+1}, \hat{O}_2^{j+1}, \dots, \hat{O}_n^{j+1}\}$ in the $(j+1)$ -th cycle;

```

1 if  $j == 0$  then
2    $\mathcal{O}^0 = \{0, 0, \dots, 0\}$ ;
3 end
4 for each link  $i$  in path do
5   Calculate the available rate  $S_i^{j+1} = W_i^{j+1} * K_i$ ;
6 end
7 Find the maximum matching rate  $S_{max}^{j+1}$ ;
8 for each link  $i$  in path do
9   Calculate the allocated capacity  $O_i^{j+1} = \lceil \frac{S_{max}^{j+1}}{K_i} \rceil$ ;
10  Sort allocated capacity of all connections in ascending order on link  $i$ ;
11   $M_i^{j+1} = C_i$ ;
12  for each connection shared on link  $i$  do
13    Calculate the actual allocated capacity  $\hat{O}_i^{j+1} = \min\{O_i^{j+1}, M_i^{j+1}\}$ ;
14     $M_i^{j+1} = M_i^{j+1} - \hat{O}_i^{j+1}$ ;
15  end
16 end
```

2) *Backward Allocation Process*: The destination hops backward propagation rate, denoted as S_{max}^{j+1} , is propagated to each repeater along the path. Ideally, each repeater can determine the allocated memory capacity on backward link i as $O_i^{j+1} = \lceil \frac{S_{max}^{j+1}}{K_i} \rceil$. The ceiling function aims to enhance the entanglement distribution rate as much as possible. It should be noted that while Alice and Bob (end nodes) only need to allocate O_i^{j+1} memory units to support a connection, all other nodes (i.e., all repeaters along the path) have to allocate O_i^{j+1} and O_{i+1}^{j+1} memory units since the repeater is shared by the two links i and $i+1$ simultaneously.

Due to asynchrony and resource contention, the theoretical allocation quantity O_i^{j+1} may not necessarily be satisfied. Considering max-min fairness guarantee and convergence, we design a sequential memory allocation mechanism based on ascending order. In this way, we can prioritize connections with poor performance, provide definite performance guarantees, and save potential link entanglement resources for serving subsequent connections with better performance. We set the spare capacity of link i as $M_i^{j+1} = C_i$ for reallocation. Then, according to Eq. (10), each repeater prioritizes satisfying the connections with the smallest allocation capacity in the current waiting queue in turn and updates the spare capacity M_i^{j+1} .

$$\hat{O}_i^{j+1} = \min\{O_i^{j+1}, M_i^{j+1}\}, M_i^{j+1} = M_i^{j+1} - \hat{O}_i^{j+1}. \quad (10)$$

It should be emphasized that both adjacent nodes of link i allocate $\hat{O}_i^{j+1}$ memory units, which can be supported by the local synchronization in link layer. The calculation of the number of allocated memory units and the memory management process in link i for the $(j+1)$ -th cycle of DRM-ETP are

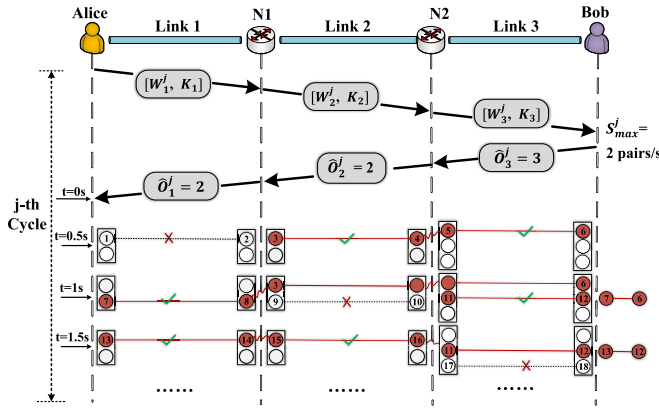


Fig. 5. The running instance in the j -th cycle of DRM-ETP on a three-hop path.

shown in Algorithm 1. Finally, if the position and number of all connections remain stable, the actual allocation of memory units after each cycle remains unchanged, although these two processes are executed periodically. Even though the link status is dynamically changing due to environmental interference, DRM-ETP can also update path information such as R_i and P_i of each link in the *Forward Seeking Process* phase to calculate the new memory units $\hat{O}_i^{j+1}$.

3) *Connection Revocation Process*: When a request has been served (e.g., sufficient entangled pairs have been distributed), source node initiates a connection revocation to release all allocated memory units on each link of the path. The released memory units can be occupied by other connections during the next several cycles through forward and backward propagation. For each connection, the revocation process only needs to be executed once and is not periodic.

To elucidate the operational dynamics of our protocol, we present a running example in Fig. 5. In the forward seeking process during j -th cycle, initiated by Alice and forwarded by N1 and N2 in sequence, the current maximum available capacity W_i^j and the entanglement generation rate K_i of one unit are aggregated to the destination. Bob calculates the maximum rate as $S_{max}^j = 2$ pairs/s according to the Eq. (9). Then, the backward allocation process is activated, initiated by Bob, and N2 and N1 sequentially transmit backward. According to Eq. (10) and Algorithm 1, the actual occupied capacity $\hat{O}_i^j$ on each link can be determined. Correspondingly, the memory units allocated on the adjacent nodes of link 1, link 2, and link 3 are assumed as 2, 2, and 3 units respectively due to different link performance. Next, on each link, continuous attempts are made to generate entangled pairs, which may succeed (✓) or fail (✗). For the convenience of explanation, we specify that the interval for attempting entanglement generation operations is 0.5s.

After the first attempt, the entanglement generation on link 1 fails. Therefore, only entanglement swapping can be completed at node N2. At this time, quantum bits 3 and 6 are entangled and wait for an entangled pair on link 1 to be created for subsequent swapping. Then, after the second attempt, the entanglement generation on link 1 is successful, and entanglement swapping is performed at node N1. At this

time, quantum bits 7 and 6 can be entangled and output for use by end users. At the same time, although an entangled pair is successfully generated on link 3, there is no adjacent entangled pair that can be matched. Finally, after the third attempt, both link 1 and link 2 successfully generate an entanglement, which can complete parallel entanglement swapping with the waiting entangled pair on link 3 and output the entangled quantum bits 13 and 12. Subsequently, quantum operations such as entanglement generation and swapping are continuously executed to distribute E2E entanglement.

D. Theoretical Analysis

1) *Optimality of Rate Matching*: Next, we provide **Theorem 1** to prove that *rate matching* can lead to the best distribution performance in the reserved path (n links) $\mathcal{P} = \{\langle R_1, C_1, P_1, O_1 \rangle, \langle R_2, C_2, P_2, O_2 \rangle, \dots, \langle R_n, C_n, P_n, O_n \rangle\}$.

Theorem 1: If the total entanglement generation rate on path $\mathcal{P}$ is T , i.e., $\sum_{i=1}^n r_i = T$ ($r_i = \frac{O_i}{C_i} R_i P_i$), when $r_i = \frac{T}{n}$, probability $\hat{P}$ is max.

Proof: As provided in Sec. IV-B, the probability $\hat{P}$ that all links on the path generate at least one entangled pair in one unit time is:

$$\begin{aligned} \hat{P} &= \prod_{i=1}^n \hat{P}_i = \prod_{i=1}^n (1 - P_i(N(t) = 0)) \\ &= \prod_{i=1}^n (1 - e^{-\frac{O_i}{C_i} R_i P_i}). \end{aligned} \quad (11)$$

Then, $\hat{P}$ is equivalently transformed into:

$$\hat{P} = \prod_{i=1}^n (1 - e^{-r_i}), \quad \sum_{i=1}^n r_i = T. \quad (12)$$

We solve the maximum value of $\hat{P}$ by *Lagrange Multiplier Method* with constructing a new function:

$$\hat{P}(r_1, \dots, r_n, \lambda) = \prod_{i=1}^n (1 - e^{-r_i}) + \lambda (\sum_{i=1}^n r_i - T). \quad (13)$$

Let the partial derivatives of each variable be 0, as follows:

$$\begin{cases} \frac{\partial \hat{P}}{\partial r_i} = e^{-r_i} \prod_{j \neq i} (1 - e^{-r_j}) + \lambda = 0. \\ \frac{\partial \hat{P}}{\partial \lambda} = \sum_{i=1}^n r_i - T = 0. \end{cases} \quad (14)$$

We calculate the solution of Eq. (14) as $r_i = \frac{T}{n}$, for $0 \leq i \leq n$. This solution equivalently shows that when the entanglement generation rate r_i allocated on each link of a connection on path $\mathcal{P}$ is the same (i.e., *rate matching*), all links generate entangled pairs simultaneously with the highest probability and are beneficial for the timely completion of subsequent swapping operations. Hence, the *rate matching* mechanism can reduce resource waste and improve distribution delay. $\square$

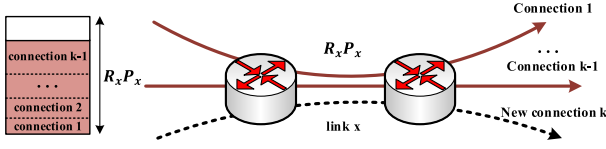


Fig. 6. A burst request connection is added to a shared link x .

2) *Analysis of Convergence and Fairness*: Assuming there are currently $k-1$ connections sharing link x with total entanglement generation rate $R_x P_x$. $k-1$ connections have reached stable rates $(S_{max}^1, S_{max}^2, \dots, S_{max}^{k-1})$, and $\sum_{i=1}^{k-1} S_{max}^i \leq R_x P_x$. Now, the k -th burst request connection starts and passes through this sharing link x , as shown in Fig. 6. At this time, there are k connections and the new average allocation rate is $\frac{R_x P_x}{k}$. Then these k connections will negotiate a new maximum rate S'_{max} in the next cycle. Suppose that among them, there are α connections whose maximum matching rate is less than or equal to $\frac{R_x P_x}{k}$, and there are $k-\alpha$ connections whose maximum matching rate is greater than $\frac{R_x P_x}{k}$. Then, we analyze processing results in the following two cases:

- Case 1: There exist α connections whose $S'_{max} \leq \frac{R_x P_x}{k}$. Since each S'_{max} does not exceed the average allocation threshold $\frac{R_x P_x}{k}$ (i.e., supply exceeds demand), it is certain that final allocated generation rate equal to S'_{max} can be allocated to these connections on link x , and $\sum_{i=1}^{\alpha} S'_{max}^i \leq \frac{\alpha}{k} R_x P_x$.
- Case 2: Besides, there also exist $k-\alpha$ connections whose $S'_{max} > \frac{R_x P_x}{k}$. Since DRM-ETP performs resource allocation in ascending order of the maximum matching rate (i.e., a connection with smaller S'_{max} has higher priority), the final allocated generation rates of these connections on link x can be expressed in turn as: $S'_{max}^{\alpha+1} = \frac{R_x P_x}{k} + \Delta_{\alpha+1}$, $S'_{max}^{\alpha+2} = \frac{R_x P_x}{k} + \Delta_{\alpha+2}$, $\dots$, $S'_{max}^k = \frac{R_x P_x}{k} + \Delta_k$, and $\Delta_{\alpha+1} \leq \Delta_{\alpha+2} \leq \dots \leq \Delta_k$. These rates satisfy the following inequalities:

$$\sum_{i=\alpha+1}^k \left(\frac{R_x P_x}{k} + \Delta_i \right) \leq R_x P_x - \sum_{i=1}^{\alpha} S'_{max}^i, \quad (15)$$

$$\sum_{i=\alpha+1}^k \Delta_i \leq \frac{\alpha}{k} R_x P_x - \sum_{i=1}^{\alpha} S'_{max}^i. \quad (16)$$

From the perspective of fairness, the dynamic rate matching mechanism based on average allocation can achieve the following goals. For connections with maximum matching rates lower than $\frac{R_x P_x}{k}$, DRM-ETP guarantees 100% resource allocation. For connections with maximum matching rates higher than $\frac{R_x P_x}{k}$, DRM-ETP assigns priority to connections that are closer to $\frac{R_x P_x}{k}$ according to the ascending order allocation principle to serve more connections passing through link x . Therefore, the final allocated generation rates S'_{max} of k connections satisfies max-min fairness.

V. PERFORMANCE EVALUATION

In this section, first, we present representative simulation scenarios and performance metrics based on comparison schemes. Second, we analyze the simulation results and provide conclusions.

A. Simulation Scenarios

We list simulation scenarios and metrics as follows:

Distribution delay: Distribution delay denotes the passed time from the initiation of the first swapping to the completion of the final entanglement distribution. Apparently, a smaller distribution delay is favorable for achieving higher-quality E2E entangled pairs.

Fidelity: Entanglement fidelity is one of the important indicators for evaluating the performance and reliability of quantum systems. To conveniently depict the negative impact of decoherence, we use Werner state entanglement [48] for modeling. Its density matrix can be expressed as: $\rho_w = w |\Phi^+\rangle\langle\Phi^+| + (1-w) \frac{I}{4}$, where $w \in [0,1]$ is a variable characterizing the content of the maximum entangled state. At this time, the corresponding fidelity of this entangled state is: $F = \frac{3w+1}{4}$. Next, the two processes of quantum memory decoherence and entanglement swapping are modeled respectively. Specifically, when a qubit is stored in quantum memory for time δt , the attenuation of its variable w conforms to $w_{delay} = w \cdot e^{-\delta t/T_{coh}}$, where w_{delay} is the parameter after decoherence, and T_{coh} is the parameter describing the decoherence rate. Entanglement swapping is another factor that leads to the decrease of entanglement fidelity. Considering that two Werner state entanglements whose corresponding parameters are w_1 and w_2 , the variable parameter of the new entanglement [49] satisfies $w_{swap} = w_1 \cdot w_2$.

Resource utilization: This is defined as the consumption ratio of link-level entangled pairs used to distribute one E2E entangled pair averagely. Each path has an optimal consumption ratio. For instance, on a n -nodes path with swapping probability Q , the optimal resource consumption ratio is $(n-1)/Q^{n-2}$. If there are invalid entanglements on the path due to prolonged waiting times, the consumption ratio exceeds the theoretical optimal value.

Throughput and fairness: The total throughput, i.e., the number of final entangled pairs, for all connections in a random network topology is computed. At the same time, we also provide the minimum value among the throughputs of each connection to reflect the service capabilities of different protocols for non-dominant connections. Subsequently, Jain's index [50] of the throughput of each connection is calculated. The Jain's index serves as a measure of the fairness of one set of data, which is widely used in data science.

$$J_{req} = \frac{(\sum_i^k t_i)^2}{k(\sum_i^k t_i^2)}, \quad (17)$$

where t_i refers to the number (or rate) of final entangled pairs of connection i , and k indicates the number of connections. J_{req} falls between $[0,1]$ with the smaller J_{req} heralding the more uneven the value of each component.

Burst request. We assess the processing performance in response to burst request in a small dumbbell topology, consistent with Fig. 4. Specifically, we introduce a sudden addition of a connection and analyze the real-time rate changes of two connections with different protocols.

We employ the open-source simulation platform SimQN [51], an advanced discrete-event-based simulation

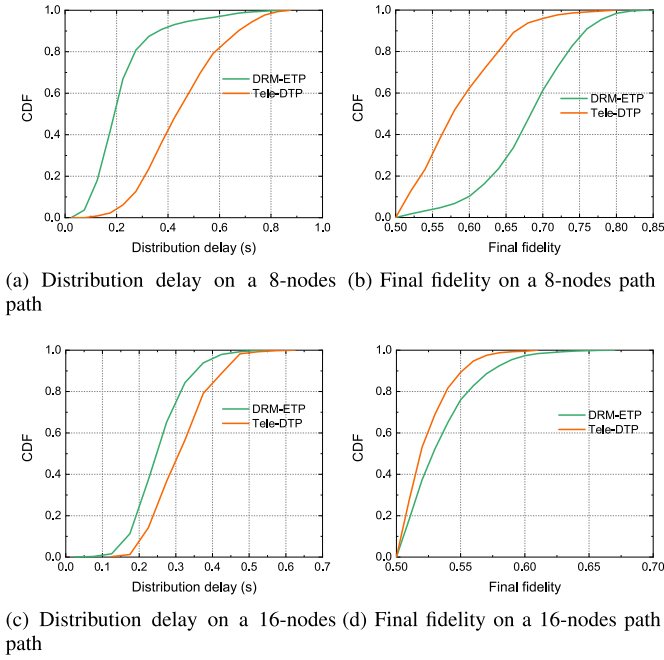


Fig. 7. Comparison of CDF with distribution delay and final fidelity on 8-nodes path and 16-nodes path.

platform for quantum networks, for our simulations. In our simulations, control and management messages, along with certain measurement results, are transmitted through a classical network (implemented in SimQN). It is assumed that all messages sent through the auxiliary classical network will be delivered reliably. We demonstrate different link performance by setting different entanglement generation rates for each link and setting different memory units for the corresponding two adjacent nodes in the simulator. To reduce accidental errors and enhance persuasiveness, in the scenarios of distribution delay, fidelity and resource utilization, we randomly construct 100 distribution paths according to the set parameters and run these protocols, and organize data into a CDF results or calculate the average value of resource consumption rate. In the scenario of throughput and fairness, we construct 100 quantum networks with either 60 or 120 nodes and randomly deploy request locations on them, where the average values of the corresponding metrics are taken as the final results.

B. Comparison Schemes

Tele-DTP. Similar to TCP, Tele-DTP [14] employs two states for each connection: slow-start and congestion avoidance. Tele-DTP incorporates a congestion encountered (CE) flag to signal the presence of memory contention along the selected path. If any node sets the CE flag, the sending window size is halved ($\lfloor w/2 \rfloor$). Tele-DTP proposes a heuristic approach akin to TCP's congestion control mechanism, utilizing both slow start and additive increase multiplicative decrease (AIMD) to regulate the quantum memory size allocated to a given connection at each quantum node.

Static Averaging (SA-ETP). Another comparison scheme is our modified version as a baseline, which eliminates the

dynamic allocation mechanism of memory units, introduced in Section IV-B. Static averaging means that each connection can be allocated up to the evenly divided memory capacity. Specifically, the available capacity, i.e., Eq. (7) is rewritten as:

$$[W_i^{j+1} = \text{threshold}_i, K_i], \quad (18)$$

which is emphasized that SA-ETP still observes *rate matching* principle. Therefore, SA-ETP is not involved in the comparison of distribution delay, fidelity and resource utilization, because these two metrics are not related to actual allocated capacity.

C. Simulation Results

1) Distribution delay and fidelity: We obtain the cumulative distribution function (CDF) of distribution delay and final fidelity on the selected paths with 8 nodes and 16 nodes, as shown in Fig. 7. We set the memory capacity of each node to random values ranging from 20 to 50, and the entanglement generation attempt rate to random values ranging from 40 times/s to 100 times/s. The probability of entanglement swapping Q and entanglement generation P are both 0.8. The maximum coherence time T_{coh} of the entangled pair is up to 1 second. We set the fidelity of the initially generated entangled pair to 0.99. Then it would experience fidelity attenuation caused by storage decoherence and entanglement swapping. When the fidelity of an entangled pair decays to below 0.5, it is considered to have lost its effectiveness and should be discarded.

When the number of nodes on the path remains unchanged, on the entanglement distribution paths of either 8 nodes or 16 nodes, DRM-ETP has lower distribution delay and higher final fidelity with using equal entanglement resources. This phenomenon verifies that the *rate matching* mechanism can significantly increase the probability that each repeater on the path has available entangled pairs in both upstream and downstream directions simultaneously. This increased probability helps complete the swapping with better synchronicity, thereby reducing E2E distribution delay and improving distribution efficiency. In addition, on paths with more nodes, more swapping operations need to be performed, and the resulting fidelity degradation is more obvious. Therefore, on the path of 16 nodes, more strict limits must be imposed on the distribution delay to meet the minimum requirements of the final fidelity higher than 0.5. So, it can be found that compared to Fig. 7(a), most of the distribution delays in Fig. 7(c) are smaller; compared to Fig. 7(b), the maximum value of the final fidelity in Fig. 7(d) decreases significantly.

2) Resource utilization: We evaluate the resource consumption ratio on paths with different numbers of nodes (ranging from 3 to 10 nodes). The entanglement generation probability P and the swapping probability Q are both set to 0.8. The remaining settings are the same as in the previous scenario. However, in practical simulations, due to the constraint of entanglement survival time, some entangled pairs exceed their survival time and are cleared due to decoherence. This leads to the actual resource consumption ratio being higher than the optimal value. We calculate the resource consumption

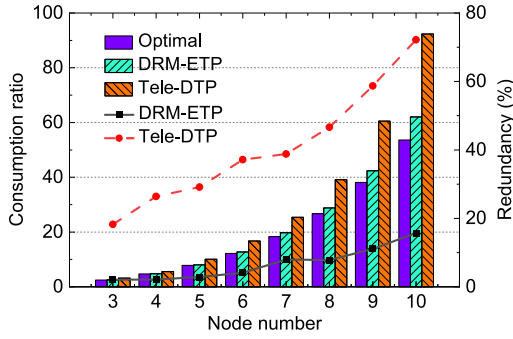


Fig. 8. Comparison of resource consumption ratio on the reserved path from 3 nodes to 10 nodes.

ratio of the two protocols in comparison with the optimal value, as depicted in Fig. 8. The left vertical axis represents the consumption ratio, while the right vertical axis illustrates the additional redundant entanglement resources consumed by the two protocols compared to the optimal value.

The evaluation reveals that with a smaller number of nodes, the disparity between the two protocols and the optimal value is negligible, as fewer entanglement swapping events do not lead to excessively long waiting times that could render entangled pairs invalid. Conversely, as the number of nodes increases, the requirement for more entanglement swapping to achieve E2E entanglement distribution becomes apparent. If the entanglement generation rate continues to be matched between each link, there is a risk of experiencing a significant waiting time with low probability. Moreover, longer paths result in more pronounced decoherence, significantly increasing the resource consumption for Tele-DTP in generating E2E entanglement.

3) Throughput and fairness: We randomly generate quantum network topologies using the Waxman [48] model in SimQN, i.e., either 60 nodes and 100 links or 120 nodes and 200 links. We set both the entanglement generation probability P and the swapping probability Q to 0.8. In the 60-nodes network, the number of connections increases from 10 to 24 in steps of 2. In the 120-nodes network, the number of connections increases from 20 to 34 in steps of 2. All connections are created at the beginning of the simulation, and the forward and backward propagation processes are completed by the source node and the subsequent entanglement distribution is executed. The node memory capacity and entanglement generation rate settings remain unchanged. In addition, we need to randomly set the locations of the source and the destination, i.e., the connection initiator and the receiver. We also ensure that the path length between the source and the destination is not less than 3 (that is, at least one entanglement swapping is required). These protocols require a pre-selected entanglement distribution path. Here, the path between the source and the destination is determined by the widely used shortest path algorithm.

By comparing the Fig. 9(a) and Fig. 9(b), the protocols based on *rate matching* exhibit higher total throughput for two main reasons: 1) The *rate matching* results in lower waiting times and less resource waste, facilitating the distribution of more entangled pairs. 2) During each forward and back-

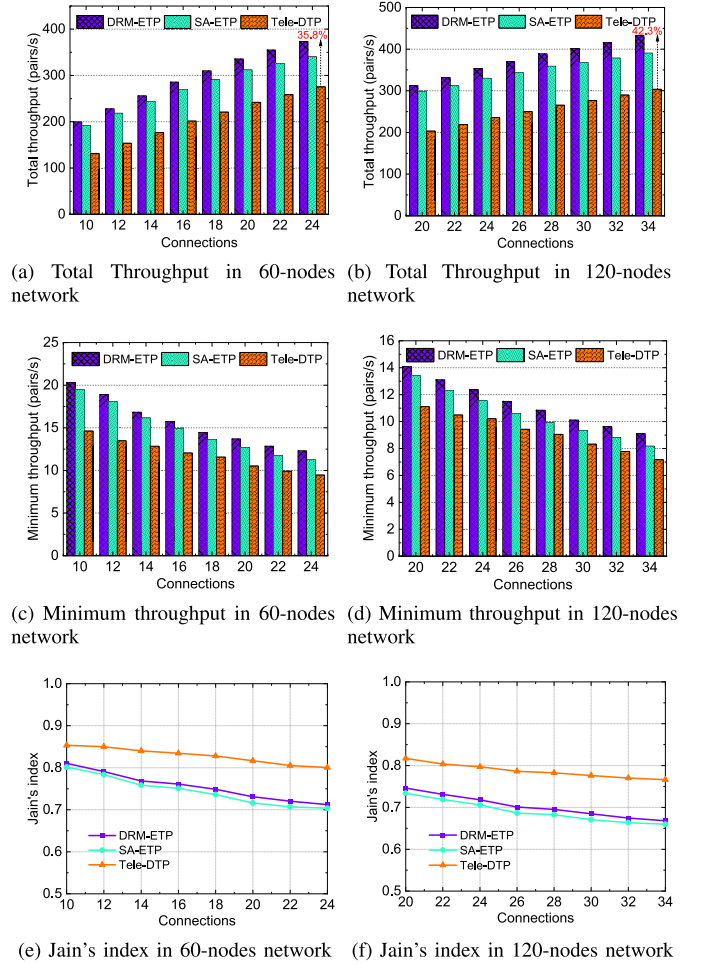


Fig. 9. Performance comparison for multiple connections in 60-nodes and 120-nodes networks respectively.

ward propagation, these protocols can leverage more refined feedback information for faster convergence when competing for limited memory with multiple connections. In contrast, the Tele-DTP protocol relies solely on the coarse-grained congestion information CE to adjust the sending window size in case of insufficient memory. Additionally, the AIMD-based window adjustment strategy in Tele-DTP slows down the convergence rate. DRM-ETP achieves better throughput than SA-ETP by fully utilizing spare memory units on shared links. As the connections increase, contention between connections intensifies, increasing the likelihood of multiple connections sharing the same link. Therefore, DRM-ETP can dynamically utilizes memory units beyond the average threshold that other connections have not occupied, leading to a widening performance gap. For example, in the two networks, when the number of connections is 24 and 34 respectively, DRM-ETP can provide a total throughput gain rate of 35.8% and 42.3% compared to Tele-DTP.

We also provide the minimum throughput and Jain's index among all connections. First, in Fig. 9(c) or Fig. 9(d), as connections increase, the minimum throughput of these three protocols continuously decreases, which is an inevitable result of intensified resource competition in shard links. Due to the DRM-ETP protocol relying on faster convergence and

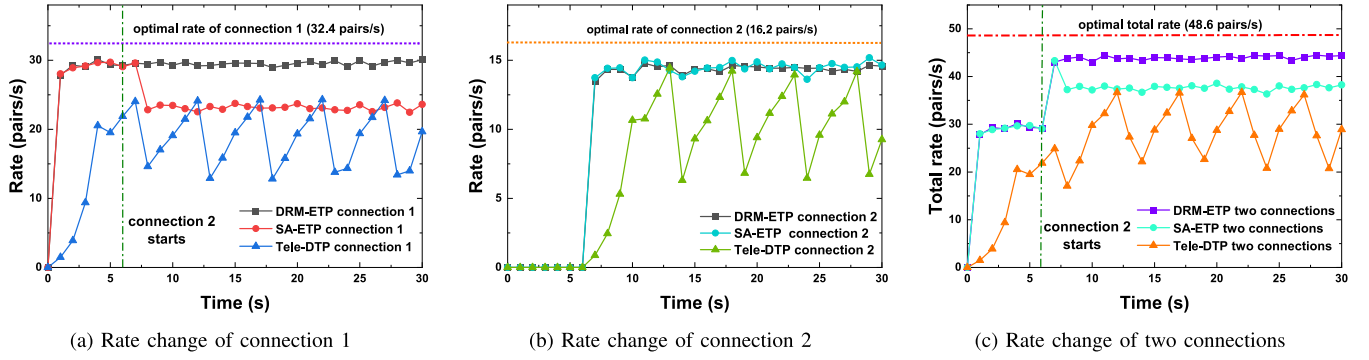


Fig. 10. Comparison of the rate change process when handling burst request connection in a dumbbell topology.

finer-grained feedback, there are fewer spare resources in the network, making its minimum throughput always perform optimally. Furthermore, although Tele-DTP demonstrates better fairness in scenarios with multiple connections, its throughput tends to be insufficient. Second, it is noteworthy that the observed fairness advantage of Tele-DTP is not highly significant, indicated by the maximum gap in Jain's index of no more than 0.1, as shown in Fig. 9(e) or Fig. 9(f). This observation can be attributed to that DRM-ETP significantly enhances the throughput of each connection because DRM-ETP maintains a higher minimum throughput and higher total throughput. Consequently, there is a slight reduction in the Jain's index. Finally, in contrast to the static averaging allocation approach employed by SA-ETP, our introduced dynamic allocation mechanism yields a marginal improvement in fairness. This enhancement arises from the mechanism to prioritize the allocation of additional memory units to non-dominant connections.

4) **Burst request:** We assess the performance of different protocols in handling burst request connection by assuming entanglement swapping with $Q = 0.9$ in a small dumbbell topology, as illustrated in Fig. 4. The memory capacity of the shared link N1-N2 is 20, while the memory capacity of other links is 10. We assume that connection 1 is initiated by user1 at the beginning, with the destination being user2, and connection 2 is initiated by user3 at the sixth second, with the destination being user4. As analyzed in Section IV, considering twice entanglement swapping, the theoretical optimal rate for connection 1 is $40 \times 0.9 \times 0.9 = 32.4$ pairs/s, and the theoretical optimal rate for connection 2 is $20 \times 0.9 \times 0.9 = 16.2$ pairs/s. We execute three protocols and calculate the real-time entanglement distribution rate for each connection during the simulation process, and the results are presented in Fig. 10.

DRM-ETP and SA-ETP both exhibit a rapid convergence, attributed to the availability of more detailed link information in the forward and backward propagation processes. On the contrary, while Tele-DTP avoids congestion, relying solely on feedback information, specifically congestion events, to control the rate proves to be slow and imprecise. The adjustment strategy of AIMD introduces significant rate fluctuations, posing challenges in maintaining a stable rate. From another perspective, by comparison of two connections, the dynamic allocation mechanism of DRM-ETP can allocate additional memory units (resulting in more link-level entangled pairs)

that cannot be utilized by connection 2 to connection 1. Upon reaching a stable rate, the rate of connection 1 in DRM-ETP is approximately 20% higher than that of connection 1 in SA-ETP.

VI. CONCLUSION

Quantum networks exhibit connection-oriented transport characteristics with continuous resource and demand models. Despite this, the existing transport protocols grapple with challenges related to strict distribution delay and adaptability to different link performance. To address these issues, we proposed a dynamic *rate matching*-based entanglement transport protocol (DRM-ETP). DRM-ETP achieves optimal *rate matching* by allocating different memory units on each link along the reserved path, accounting for different link performance. Additionally, DRM-ETP introduces a dynamic allocation mechanism based on explicit average allocation to handle burst requests, thereby enhancing performance and ensuring fairness. Through extensive simulations, we demonstrated that DRM-ETP outperforms the existing entanglement transport protocol Tele-DTP, distributing more long-distance entangled pairs with less delay and improved efficiency, especially in network systems with obvious differences in link performance and distributed mechanism. Its dynamic and fine-grained control enables closer-to-optimal rates and fast convergence when handling burst requests. Our research opens avenues for more effective and adaptable entanglement distribution in quantum networks, bringing us closer to the realization of advanced quantum applications.

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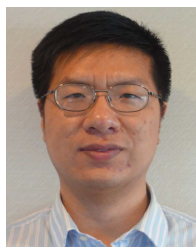


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